# 14.1 The partial derivative



Let $z = f(x,y)$ be a function and $(a,b)$ a point.

$$f_x(a,b) = \text{rate of change of } f \text{ in the positive x-direction} = \lim_{h \to 0} \frac{f(a+h,b) - f(a,b)}{h}$$

$$f_y(a,b) = \text{rate of change of } f \text{ in the positive y-direction} = \lim_{h \to 0} \frac{f(a,b+h) - f(a,b)}{h}$$

These are called partial derivatives. Alternate notations:

$$f_x(x,y) = \frac{\partial f}{\partial x} \text{ or } \frac{\partial z}{\partial x} \qquad f_y(x, \ ) = \frac{\partial f}{\partial y} \text{ or } \frac{\partial z}{\partial y}$$

→ Do all problems.